D5. Discovering groups: clustering

Unsupervised learning: there is no outcome. Clusters depend on the variables, the scaling, the method and k.

1. Purpose and variables

Write the question, e.g. "Are there distinct lifestyle profiles?"

Choose variables for substantive reasons. Every variable you add changes the distances.

Leave out outcomes you want to compare later (e.g. use activity, diet and BMI; keep SBP aside).

2. Prepare the data

Complete cases: `kmeans()` cannot handle missing values.

Very skewed variables may need a log transformation.

Variables on different scales or units?
e.g. minutes (0 to 900) and a score (0 to 10)

Yes (usual): standardise each variable to mean 0, SD 1

```
z <- scale(life[,
c("activity_min",
    "diet_score", "bmi")])
```

Without scaling, the variable with the largest numbers dominates.

3. Choose a method

k-means
Numeric variables; you set k in advance; finds compact, roughly round groups. Use several random starts.

```
set.seed(2024)
kmeans(z, centers = 3,
       nstart = 25)
```

Hierarchical clustering
Builds a tree (dendrogram) by merging the closest groups; cut it to get k groups. Choose a distance and a linkage (Ward's method is common).

```
hc <- hclust(dist(z),
     method = "ward.D2")
cutree(hc, k = 3)
```

4. Choose the number of clusters

Elbow plot: within-cluster sum of squares against k; look for a bend.

Average silhouette width: how well points sit in their own cluster (maximum 1).

Interpretability: can you describe and use the groups?

There is often no single right k. In the course data the elbow is gradual and all silhouettes are below 0.35: weak structure. Say so.

5. Check and describe

Describe clusters on the **original** scale (medians, means, n).

Check stability: other seeds, k, methods or subsets.

Compare clusters on variables **not** used to build them (e.g. age, SBP).

Apparent clusters are not automatically real. k-means always returns k groups, even from data with no groups. Do not test whether clusters differ on the variables used to create them.

Further learning: mixed numeric and categorical variables (Gower distance with `cluster::daisy` and `pam`); model-based clustering (Gaussian mixture models); principal component analysis for visualising many variables.